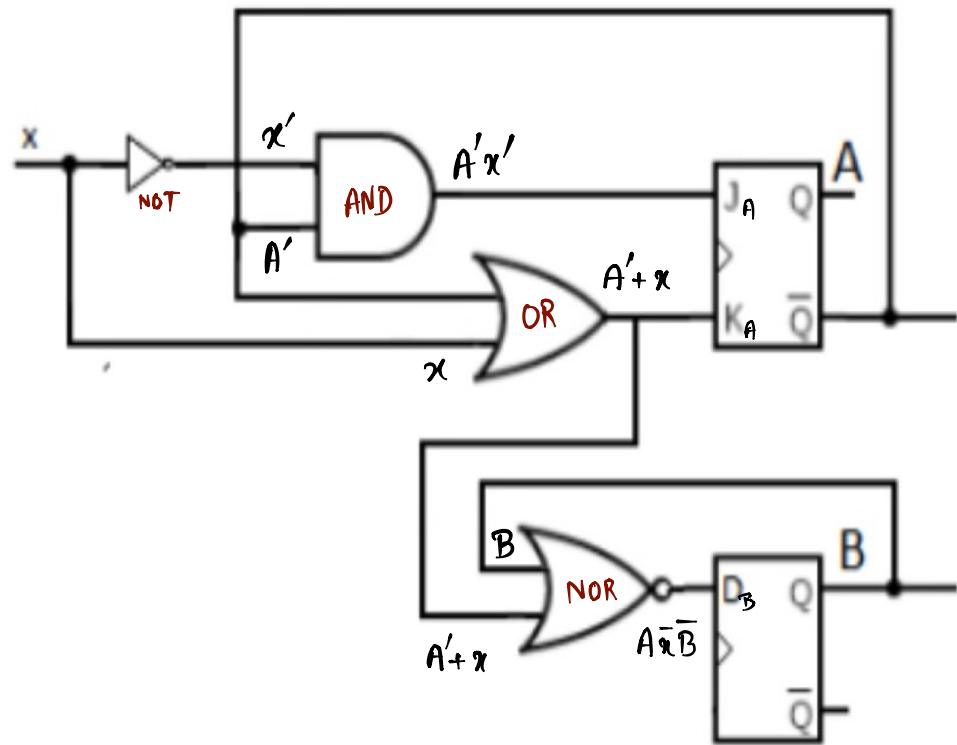


1. [1.5+4+2] Draw the state diagram for the given circuit.



No of flip-flop = 2

Present States : A, B

Next States : A^+ , B^+

Input : J_A , K_A , D_B , x

Equations :

$$J_A = A'x'$$

$$K_A = A' + x$$

$$D_B = (A' + x + B)' = A''x'B' = Ax'B'$$

J-K Flip-flop Characteristic Table

Q	J	K	Q^+
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	1
1	0	0	1
1	0	1	0
1	1	0	1
1	1	1	0

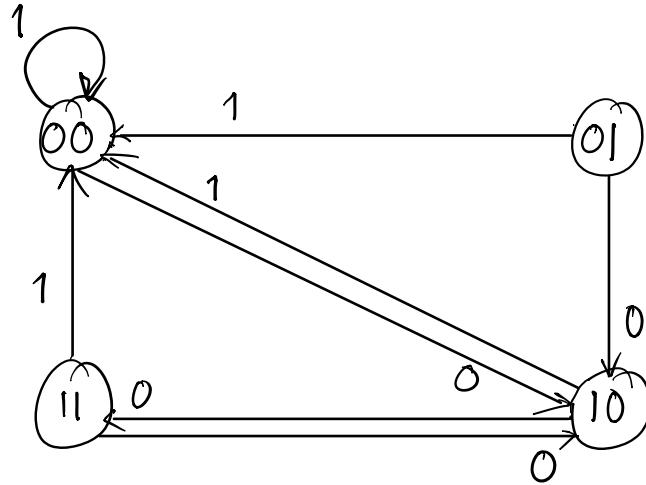
D Flip-flop Characteristic Table

Q	D	Q^+
0	0	0
0	1	1
1	0	0
1	1	1

State Table :

A	B	X	J _A	K _A	D _B	A ⁺	B ⁺
0	0	0	1	1	0	1	0
0	0	1	0	1	0	0	0
0	1	0	1	1	0	1	0
0	1	1	0	1	0	0	0
1	0	0	0	0	1	1	1
1	0	1	0	1	0	0	0
1	1	0	0	0	0	1	0
1	1	1	0	1	0	0	0

State Diagram :



2. [3+1+1+2.5] Implement the following counter using T FF:

Green->Orange->Yellow->Red->Yellow->Orange->Yellow->Green

AB: 00 → 01 → 10 → 11 → 10 → 01 → 10 → 00
 xy : 00 01 10

For the combination $x=1$ &

$y=1$, I am using 'going to the initial state'. You can use 'Don't care' as well, as there is nothing explicitly mentioned in the question.

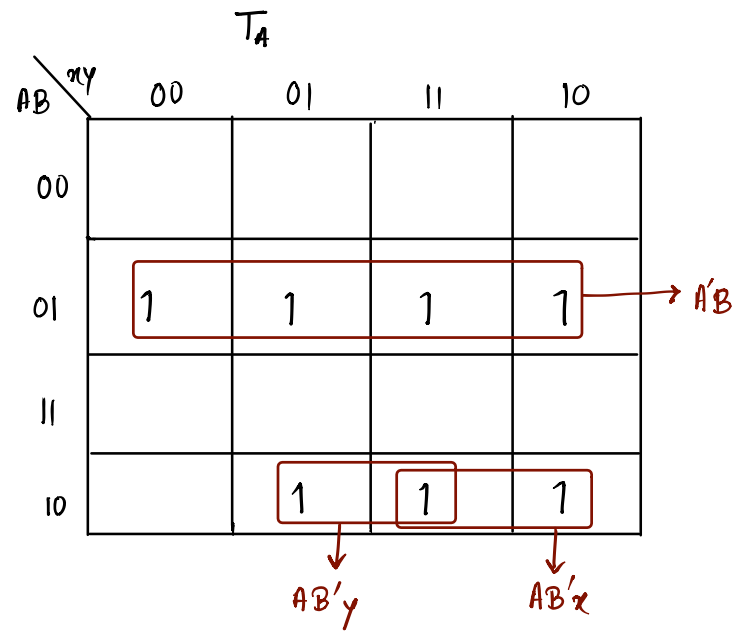
T Flip-flop Excitation Table

Q	Q ⁺	T
0	0	0
0	1	1
1	0	1
1	1	0

State Table :

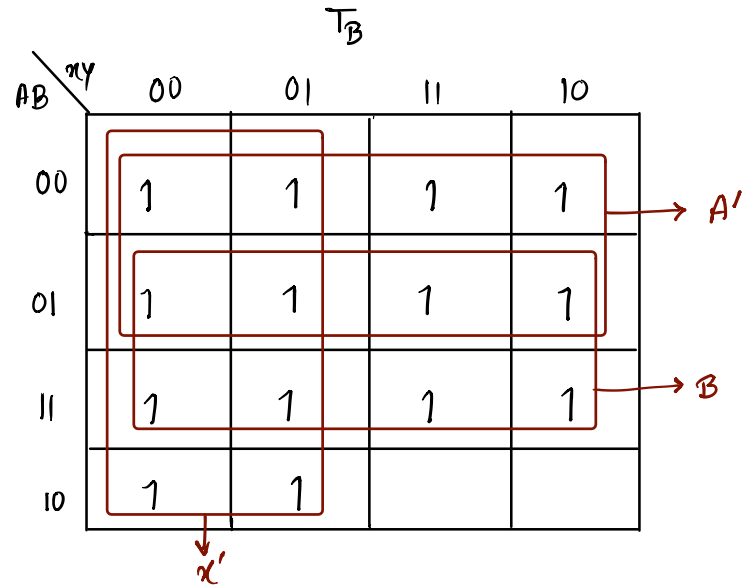
A	B	x	y	A ⁺	B ⁺	T _A	T _B
0	0	0	0	0	1	0	1
0	0	0	1	0	1	0	1
0	0	1	0	0	1	0	1
0	0	1	1	0	1	0	1
0	1	0	0	1	0	1	1
0	1	0	1	1	0	1	1
0	1	1	0	1	0	1	1
0	1	1	1	1	0	1	1
1	0	0	0	1	1	0	1
1	0	0	1	0	1	1	1
1	0	1	0	0	0	1	0
1	0	1	1	0	0	1	0
1	1	0	0	1	0	0	1
1	1	0	1	1	0	0	1
1	1	1	0	1	0	0	1
1	1	1	1	1	0	0	1

K-Maps of Flip-flop Inputs :



$$T_A = A'B + AB'y + AB'x$$

$$= A'B + AB'(x+y)$$



$$T_B = A' + B + x'$$

Circuit Diagram :

